GeoEnrich-NYC: A Geospatial Text-Tabular Dataset for Airbnb Price Prediction

Ely Raz
ely.raz@campus.technion.ac.il

Abstract

We present GeoEnrich-NYC, a text-tabular dataset of 14,990 New York City Airbnb listings for nightly-price regression, in which each listing's coordinates are replaced by an LLM-generated description of its surroundings, built from Overture Maps places data. Evaluated with the MulTaBench curation pipeline, the dataset passes both curation criteria on all five learners: adding the description to the structured columns lowers RMSE on log price by 0.059–0.075, and fine-tuning the text encoder on the target lowers it by a further 0.002–0.007. The enrichment pipeline can be applied to other datasets with latitude-longitude coordinates, though we evaluate it on a single city.

1 Introduction

Accommodation datasets (Airbnb, Booking.com) often include latitude-longitude coordinates of listings. The location of a listing can be leveraged as a significant indicator for price prediction regression tasks, though the coordinates themselves say little about what surrounds a listing. To solve this problem, traditional ML regression models engineer location features (e.g., spatial clustering) to leverage this potentially informative feature.

In addition, the surroundings of a listing can often provide signal regarding its price. For example: It is reasonable to assume [Rosen, 1974, Wang and Nicolau, 2017] that Airbnb listings in downtown Manhattan, where there is a high density of restaurants and tourist attractions, will be pricier than listings in the outskirts of the city—where few or no amenities can be found.

We therefore hypothesize that a natural-language description of a listing's surroundings will improve models' fit in a nightly-price prediction regression task.

To achieve this, we leverage the given latitude-longitude coordinates from an existing Airbnb listings dataset and enrich them with geospatial data by querying Overture Maps places data and enriching each listing with business density information of its surroundings. We then use an LLM to transform the numerical density information into a description of each listing's surroundings.

Our goal is a dataset that passes the MulTaBench [Arazi et al., 2026] text-tabular curation criteria, which GeoEnrich-NYC does on all five learners (Section 4).

2 Related work

Text-tabular curation. MulTaBench [Arazi et al., 2026] admits a text-tabular dataset only if, across a committee of tabular learners, joint frozen text-tabular inputs improve on both unimodal baselines, and target-aware representations (TAR), as in TabSTAR [Arazi et al., 2025], improve on frozen ones. We follow its protocol, with the one adaptation described in Section 4. Earlier text-tabular benchmarks include Airbnb price datasets [Shi et al., 2021]; MulTaBench's curation rejected both Airbnb datasets it evaluated (Melbourne Airbnb, 1/5 learners; Airbnb Seattle, 0/5).

LLM enrichment. Large language models are increasingly used as components of data pipelines, not only as predictors. Previous studies have leveraged modern LLMs to generate a natural-language description of tabular features [Hegselmann et al., 2023]. Closest to our pipeline, GeoLLM [Manvi et al., 2024] prompts an LLM with coordinates and nearby OpenStreetMap places to predict location-level indicators, including home values. We became aware of it only after building our pipeline, which differs in withholding the coordinates from the LLM and leaving prediction to tabular learners.

3 Dataset origins and curation methodology

3.1 Dataset and sources

Inside Airbnb is a project that provides quarterly data and advocacy about Airbnb's impact on residential communities; its data is freely available online under a CC BY 4.0 licence. We used their New York City listings snapshot [Inside Airbnb, 2026], scraped 14–23 June 2026.

Overture Maps Foundation is a non-profit organization that provides free and open-source maps data. We used DuckDB to query the places theme (licensed under CDLA-Permissive-2.0) of Overture release `2026-08-19.0` [Overture Maps Foundation, 2026] for businesses and points of interest in our desired city, and performed a join by proximity for each listing, with distances measured in a metric projection (UTM zone 18N).

3.2 Pipeline

Data cleaning. After acquiring the original dataset from "Inside Airbnb", we perform a standard feature engineering pipeline: columns that do not partake in the regression task are dropped, the schema is defined and enforced, and features such as "number of rooms" and "number of baths" are extracted from semi-structured text fields using simple regex rules. Listings with missing or non-positive prices are dropped, as are listings missing coordinates or more than 25% of their features. In addition, the target nightly-price column of the final version of the dataset is log-transformed: $\text{target} = \log(1 + \text{price})$. Of the 30,259 listings in the snapshot, 21,155 survive these filters (Table 3); we randomly sample 15,000 of them (seed 0), each with 8 feature columns (six numerical, one boolean, and one categorical) besides the target. The schema and feature distributions are summarized in Table 2 (Appendix A).

Geospatial enrichment. We query Overture for all places within the listings' bounding box (padded by $\approx 2\,\text{km}$) that have a confidence score of at least 0.6 and belong to one of our selected categories. We apply a join such that each listing coordinate is assigned all POIs (points of interest) in a 450-meter radius (roughly 570 meters of walking on the Manhattan street grid), saved in JSON format. The 10 listings with no POI within that radius are dropped, leaving 14,990 listings. The original latitude-longitude columns are dropped by design: location reaches the models only through the text column, so the structured columns carry no location information.

Aggregation. Each JSON column describing nearby POIs is reduced down to a simple count, and the distance to the nearest place, according to a list of 20 predefined buckets (for example: restaurants, parks, ground transport, and nightlife venues). Those categories are curated from the level-2 categories of the official Overture hierarchical taxonomy. We rank each listing's count in every category against the citywide distribution of that count over all listings, and replace the count-buckets of each listing with its relative percentile. The ranking uses no price information, but is computed over all listings, including those later used for testing. Popular POIs (landmarks)—77 curated attractions such as the Empire State Building, the Statue of Liberty, and Times Square—are matched within 800 meters of each listing and not reduced down to their categories, as they potentially carry stronger neighbourhood signal, which is also why they are matched over a wider radius.

LLM descriptive summary. By the end of the aggregation stage, each row has a JSON object containing the listing's relative density of each category in its 450-meter surrounding radius and a list of popular tourist attractions nearby, including their exact distance. We take a prompt blueprint and structure the JSON summaries, preparing them for an LLM API call: percentiles are phrased as word bands (e.g., "far more than most neighbourhoods"), and only the three nearest landmarks are kept, with their distance phrased as a band (e.g., "a short walk from"). We send batch requests to an LLM

(1) Measured, with citywide rank	(2) Evidence line sent to the LLM
259 places within 450 m, 60th percentile	`Overall: busy, by how many places...`
4 discount stores, 97th percentile	`discount stores: far more than most neighbourhoods`
nearest historic site at 53 m ($\leq$ 150 m)	`historic sites: steps away`
Central Park at 792 m ($>$ 500 m)	`about 10 minutes from: Central Park`
e.g. 83 restaurants, 59th percentile	*16 categories within the typical range: omitted*

(3) Generated `surroundings_summary`: "This busy neighbourhood features far more discount stores than most neighbourhoods, with historic sites steps away. Central Park is about 10 minutes away."

Figure 1: How listing 10433 of the released dataset (entire home, 7 guests) gets its description. Each row traces one piece of evidence; colours mark where it surfaces in the summary. Of the 18 place categories near this listing, only one is a citywide contrast (outer 15%); typical ones are omitted. Only the lines in (2), in a per-listing shuffled order, reach the LLM; the measurements in (1) do not.

(temperature 0.4), processing all 14,990 rows independently in three batches. The LLM describes the surroundings and is not given the coordinates, exact counts, exact distances, or listing price, to prevent leakage. Figure 1 traces one listing through these steps.

Multiple LLMs were tested on subsets of the dataset; Gemini 3.5 Flash-Lite [Google DeepMind, 2026] was chosen for its relative lack of hallucinations on those subsets, generation speed, batch-request availability, and lack of reasoning. The prompt was refined over 16 variants, tested on a subset of $\approx$ 2,000 rows, before the final variant was fixed.

The released dataset thus pairs the structured columns with a single text column, `surroundings_summary` (statistics in Appendix C). The code, prompts, word-band rules, and landmark list are available at `https://github.com/razilai/GeoEnrich`. The dataset is available on Hugging Face ([link]) in MulTaBench's format (typed `data.parquet` and `metadata.json`) under CC BY 4.0, attributing Inside Airbnb and the Overture Maps Foundation.

4 Results

With the prompt fixed, we evaluate GeoEnrich-NYC once with the MulTaBench benchmark, run from its current repository (`master` at commit `d88821d`) with no change to its evaluation code; the dataset was added to its registry and served from a local Hugging Face cache. The curation committee consists of five learners: two GBDTs, LightGBM [Ke et al., 2017] and CatBoost [Prokhorenkova et al., 2018], the MLP ensemble TabM [Gorishniy et al., 2025], and the tabular foundation models TabPFNv2 [Hollmann et al., 2025] and TabPFNv2.5 [Grinsztajn et al., 2025]. Each is trained under four conditions: structured columns only, the surroundings description only (text-only), structured columns with frozen E5-small [Wang et al., 2022] text embeddings (joint frozen), and structured columns with E5-small embeddings fine-tuned on the target with low-rank adaptation [LoRA; Hu et al., 2022] (joint TAR). Every condition runs on the benchmark's six splits of TabArena's [Erickson et al., 2025] 67/33 scheme ($\approx$ 9,993 training and $\approx$ 4,997 test rows per split). The metric is the RMSE on the log-price scale, $\log(1 + \text{price})$, as reported by the current repository (the MulTaBench paper uses R^2); lower is better.

A learner shows *joint signal* when its mean joint-frozen RMSE is lower than both its structured and its text-only RMSE by more than 0.001, and *TAR gain* when its mean joint-TAR RMSE is lower than its joint-frozen RMSE by more than 0.001 (means rounded to three decimals before differencing). The dataset is eligible when at least 3 of the 5 learners show both. We adapt the acceptance criterion of Arazi et al. [2026, App. A.3] to RMSE rather than R^2; since the current repository no longer includes it, we compute it with our own code.

Table 1 reports the results. Adding the surroundings description to the structured columns lowers RMSE for every learner, by 0.059–0.075 (11.8–15.3% relative to the structured baseline), so all five learners show joint signal. The description alone is a weak predictor: text-only RMSE (0.710–0.726) is well above the structured baseline for every learner, which suggests the text complements the listing attributes, supplying the location information they lack by design, rather than substituting for them. Fine-tuning the encoder on the target lowers RMSE further for all five learners, so all five show

Table 1: RMSE on $\log(1 + \text{price})$, mean $\pm$ standard deviation over six splits (lower is better). Δ_{joint}: improvement of joint frozen over the better unimodal baseline. Δ_{TAR}: improvement of joint TAR over joint frozen.

Learner	Structured	Text-only	Joint frozen	Joint TAR	Δ_{joint}	Δ_{TAR}
LightGBM	0.501 ± 0.007	0.726 ± 0.006	0.442 ± 0.006	0.435 ± 0.005	+0.059	+0.007
CatBoost	0.498 ± 0.006	0.722 ± 0.008	0.435 ± 0.005	0.431 ± 0.007	+0.063	+0.004
TabM	0.505 ± 0.008	0.720 ± 0.009	0.434 ± 0.004	0.432 ± 0.005	+0.071	+0.002
TabPFNv2	0.490 ± 0.008	0.715 ± 0.007	0.422 ± 0.005	0.418 ± 0.006	+0.068	+0.004
TabPFNv2.5	0.490 ± 0.007	0.710 ± 0.008	0.415 ± 0.004	0.412 ± 0.005	+0.075	+0.003

TAR gain and the dataset passes the MulTaBench curation criterion with 5/5 learners on both checks. The TAR gain is small, however (0.002–0.007), and of the same order as the split-to-split standard deviation; joint TAR beats joint frozen on 27 of the 30 learner–split pairs (6/6 splits for LightGBM, TabPFNv2 and TabPFNv2.5; 5/6 for CatBoost; 4/6 for TabM; per-learner paired differences in Appendix B). The best configuration, TabPFNv2.5 with joint TAR, reaches an RMSE of 0.412, a 15.9% reduction over its structured baseline.

5 Conclusion, limitations, and future work

5.1 Outcomes

GeoEnrich-NYC passes both MulTaBench curation criteria on all five learners. Most of the gain comes from adding the surroundings description to the structured columns; the further gain from fine-tuning the encoder is an order of magnitude smaller and comparable to split-to-split variation.

5.2 Limitations and future work

Evaluation. It is unclear what the main driver of the price signal is in each listing's surroundings description. It is possible that it is derived from the explicit naming of popular landmarks (present in the evidence of 42.6% of listings), or perhaps by the overall busyness band and the category contrasts. The pipeline already builds two censored arms that separate these sources — one keeping only the landmark names found in each summary, and one replacing each landmark name with "a well-known landmark" — but they were not evaluated in this release. The prompt was selected on rows that are also part of the released dataset, and was not re-validated on held-out rows. MulTaBench's random splits also place nearby listings, which share near-identical surroundings, in both training and test sets (12,594 distinct rendered evidence blocks for 14,990 listings), which may inflate the gains relative to unseen areas. Finally, Overture's places coverage is likely uneven across the city, so a "quiet" band may partly reflect sparse data.

Application. The pipeline needs only latitude-longitude coordinates, so it can be applied to other datasets, including Inside Airbnb's other cities, but we have evaluated it on a single city. The prompts are worded for hospitality and real-estate tasks, and the manually curated landmark list must be rebuilt for each city.

Future work. Future work could evaluate the pipeline beyond hospitality price regression. Since coordinates alone could carry the same or a stronger location signal, it could also rerun the four conditions with coordinates kept as structured columns and test whether TAR still beats joint frozen.

References

Alan Arazi, Eilam Shapira, and Roi Reichart. TabSTAR: A tabular foundation model for tabular data with text fields. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38, 2025.

Alan Arazi, Eilam Shapira, Shoham Grunblat, Mor Ventura, Elad Hoffer, Gioia Blayer, David Holzmüller, Lennart Purucker, Gaël Varoquaux, Frank Hutter, and Roi Reichart. MulTaBench: Benchmarking multimodal tabular learning with text and image. *arXiv preprint arXiv:2605.10616*, 2026.

Nick Erickson, Lennart Purucker, Andrej Tschalzev, David Holzmüller, Prateek Mutalik Desai, David Salinas, and Frank Hutter. TabArena: A living benchmark for machine learning on tabular data. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2025.

Google DeepMind. Gemini 3.5 Flash-Lite model card. `https://deepmind.google/models/model-cards/gemini-3-5-flash-lite/`, 2026.

Yury Gorishniy, Akim Kotelnikov, and Artem Babenko. TabM: Advancing tabular deep learning with parameter-efficient ensembling. In *International Conference on Learning Representations (ICLR)*, 2025.

Léo Grinsztajn, Klemens Flöge, Oscar Key, Felix Birkel, Philipp Jund, Brendan Roof, Benjamin Jäger, Dominik Safaric, Simone Alessi, Adrian Hayler, et al. TabPFN-2.5: Advancing the state of the art in tabular foundation models. *arXiv preprint arXiv:2511.08667*, 2025.

Stefan Hegselmann, Alejandro Buendia, Hunter Lang, Monica Agrawal, Xiaoyi Jiang, and David Sontag. TabLLM: Few-shot classification of tabular data with large language models. In *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 206 of *Proceedings of Machine Learning Research*, pages 5549–5581, 2023.

Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637(8045):319–326, 2025. doi: 10.1038/s41586-024-08328-6.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.

Inside Airbnb. New York City listings, snapshot of 14 June 2026. `https://data.insideairbnb.com/united-states/ny/new-york-city/2026-06-14/data/listings.csv.gz`, 2026.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

Rohin Manvi, Samar Khanna, Gengchen Mai, Marshall Burke, David Lobell, and Stefano Ermon. GeoLLM: Extracting geospatial knowledge from large language models. In *International Conference on Learning Representations (ICLR)*, 2024.

Overture Maps Foundation. Overture Maps places theme, release 2026-08-19.0. `https://docs.overturemaps.org/guides/places/`, 2026.

Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

Sherwin Rosen. Hedonic prices and implicit markets: Product differentiation in pure competition. *Journal of Political Economy*, 82(1):34–55, 1974. doi: 10.1086/260169.

Xingjian Shi, Jonas Mueller, Nick Erickson, Mu Li, and Alexander J. Smola. Benchmarking multimodal AutoML for tabular data with text fields. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2021.

Dan Wang and Juan L. Nicolau. Price determinants of sharing economy based accommodation rental: A study of listings from 33 cities on Airbnb.com. *International Journal of Hospitality Management*, 62:120–131, 2017. doi: 10.1016/j.ijhm.2016.12.007.

Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533*, 2022.

A Data card

B Paired TAR gains

Table 2: Schema and feature distributions of the released dataset (14,990 rows). Quantiles are over non-missing values. The target is $\log(1 + \text{price})$, with price the nightly price in USD (raw quantiles: 7.2, 96, 173, 299, 30,973).

Column	Type	Missing	Min	Q1	Median	Q3	Max
`price` (target)	log USD	0.0%	2.10	4.58	5.16	5.70	10.34
`ratings`	numeric	27.8%	1.00	4.67	4.86	5.00	5.00
`guests`	numeric	0.0%	1	2	2	4	16
`bathrooms`	numeric	3.6%	0	1	1	1	15.5
`num_bedrooms`	numeric	33.8%	0	1	1	2	19
`property_number_of_reviews`	numeric	0.0%	0	0	6	41	4502
`minimum_nights`	numeric	0.0%	1	30	30	30	365
`room_type`	categorical	0.0%	Entire home 54.7%, Private 42.1%, Hotel 2.3%, Shared 0.9%				
`is_superhost`	boolean	0.0%	True 31.1%				
`surroundings_summary`	text	0.0%	14,254 distinct				

Table 3: Row attrition from the Inside Airbnb snapshot to the released dataset, and duplicates.

Step	Rows
NYC snapshot (scraped 14–23 June 2026)	30,259
Positive, non-missing price	21,515
Coordinates present and at most 25% of features missing	21,155
Random sample (seed 0)	15,000
At least one POI within 450 m (released)	14,990
Rows whose 8 tabular features and summary duplicate another row	46

Table 4: Per-learner paired difference between joint frozen and joint TAR RMSE on $\log(1 + \text{price})$, computed split by split over the six splits (positive means TAR is better).

Learner	Paired difference, mean $\pm$ SD	Splits won by TAR
LightGBM	0.0074 ± 0.0024	6/6
CatBoost	0.0047 ± 0.0032	5/6
TabM	0.0014 ± 0.0033	4/6
TabPFNv2	0.0035 ± 0.0017	6/6
TabPFNv2.5	0.0036 ± 0.0010	6/6

C Summary text statistics

Table 5: Statistics of `surroundings_summary` over the 14,990 released rows. Words are alphabetic tokens. Jaccard similarity is computed on content-word sets (stopwords and generic words such as "area" and "neighbourhood" removed) over 20,000 random pairs (seed 0); lower means more varied text.

Statistic	Value
Words per summary, mean ± SD (median; min–max)	39.1 ± 11.1 (38; 11–84)
Characters per summary, mean (median; max)	238.7 (229; 522)
Sentences per summary: 1 / 2 / 3	450 / 14,071 / 469
Distinct summaries	14,254 (95.1%)
Distinct rendered evidence blocks	12,594 (84.0%)
Vocabulary (lower-cased word types)	442
Mean pairwise content-word Jaccard similarity	0.185
Listings with a landmark in the evidence	42.6%
of which the summary names at least one landmark	99.6%
Summaries containing "quiet" / "busy"	39.4% / 39.9%